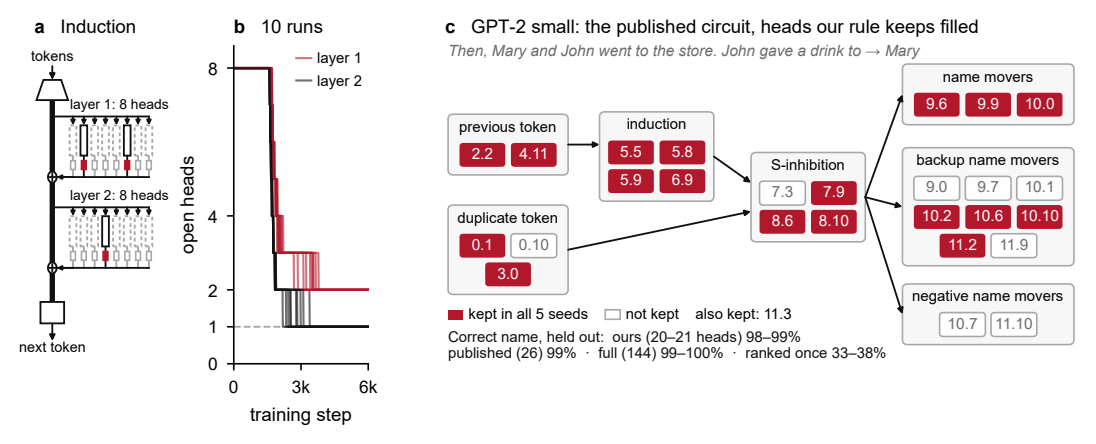

a Induction
tokens
layer 1: 8 heads
layer 2: 8 heads
next token
b 10 runs
layer 1
layer 2
open heads
8
4
2
1
0
0
3k
6k
training step
c GPT-2 small: the published circuit, heads our rule keeps filled
Then, Mary and John went to the store. John gave a drink to → Mary
previous token
2.2
4.11
induction
5.5
5.8
5.9
6.9
duplicate token
0.1
0.10
3.0
S-inhibition
7.3
7.9
8.6
8.10
name movers
9.6
9.9
10.0
backup name movers
9.0
9.7
10.1
10.2
10.6
10.10
11.2
11.9
negative name movers
10.7
11.10
kept in all 5 seeds
not kept
also kept: 11.3
Correct name, held out: ours (20–21 heads) 98–99%
published (26) 99% · full (144) 99–100% · ranked once 33–38%